



HACKATHON PITCH PRESENTATION

AI Fitness Assistant

Real-Time Computer Vision Pose Tracking & Conversational AI
Dietician

CORE TECH

MediaPipe CV • OpenCV • Streamlit • LLM

TARGET

Democratized Personalized Personal Training

THE CORE PROBLEM

The Accessibility Gap in Personal Fitness Coaching

\$ Prohibitive Cost Barrier

Personal trainers cost \$50–\$150/hour, making human guidance an unaffordable luxury for most.

⚠ High Injury Risk from Bad Form

Solo exercisers suffer injuries and stalled progress due to uncorrected posture mistakes.

80%

Of solo gym beginners quit within 90 days due to lack of guidance and injuries.

Three Major Pillars Breaking the User Journey



Zero Form Verification

Fitness apps show static videos but cannot correct dynamic user movement in real-time.



Generic Nutrition Advice

Standard meal plans fail to address real-time user dietary questions and post-workout needs.



Manual Rep Logging

Manual workout tracking creates friction, breaking exercise momentum and data accuracy.

OUR SOLUTION

AI Fitness Assistant: Computer Vision + Conversational AI

1 Vision-Based Trainer

Tracks 33 body keypoints via standard webcam for posture precision.

2 Automatic Rep & Calorie Counter

State-machine logic counts completed reps and computes energy burn.

3 AI Dietician Chatbot

Instant LLM-powered nutrition advice tailored to workout goals.



Real-Time Computer Vision Engine

MEDIA PIPE POSE

33 Landmark Joint Tracking

Extracts 3D coordinates for shoulders, elbows, hips, knees, and ankles in real-time.

TRIGONOMETRIC GEOMETRY

Dynamic Angle Calculation

Uses vector arctangent math to evaluate joint angle thresholds per movement phase.

`</>` Mathematical Pose Verification

$$\theta = \arctan2(y3-y2, x3-x2) - \arctan2(y1-y2, x1-x2)$$

If Bicep Curl angle exceeds 160° → Stage set to DOWN.

If Angle contracts below 30° → Stage set to UP & Rep ++.

Multi-Exercise Pose Recognition & Evaluation



Bicep Curls

Elbow extension & contraction tracking.



Squats & Lunges

Knee angle & hip depth verification.



Pushups & Planks

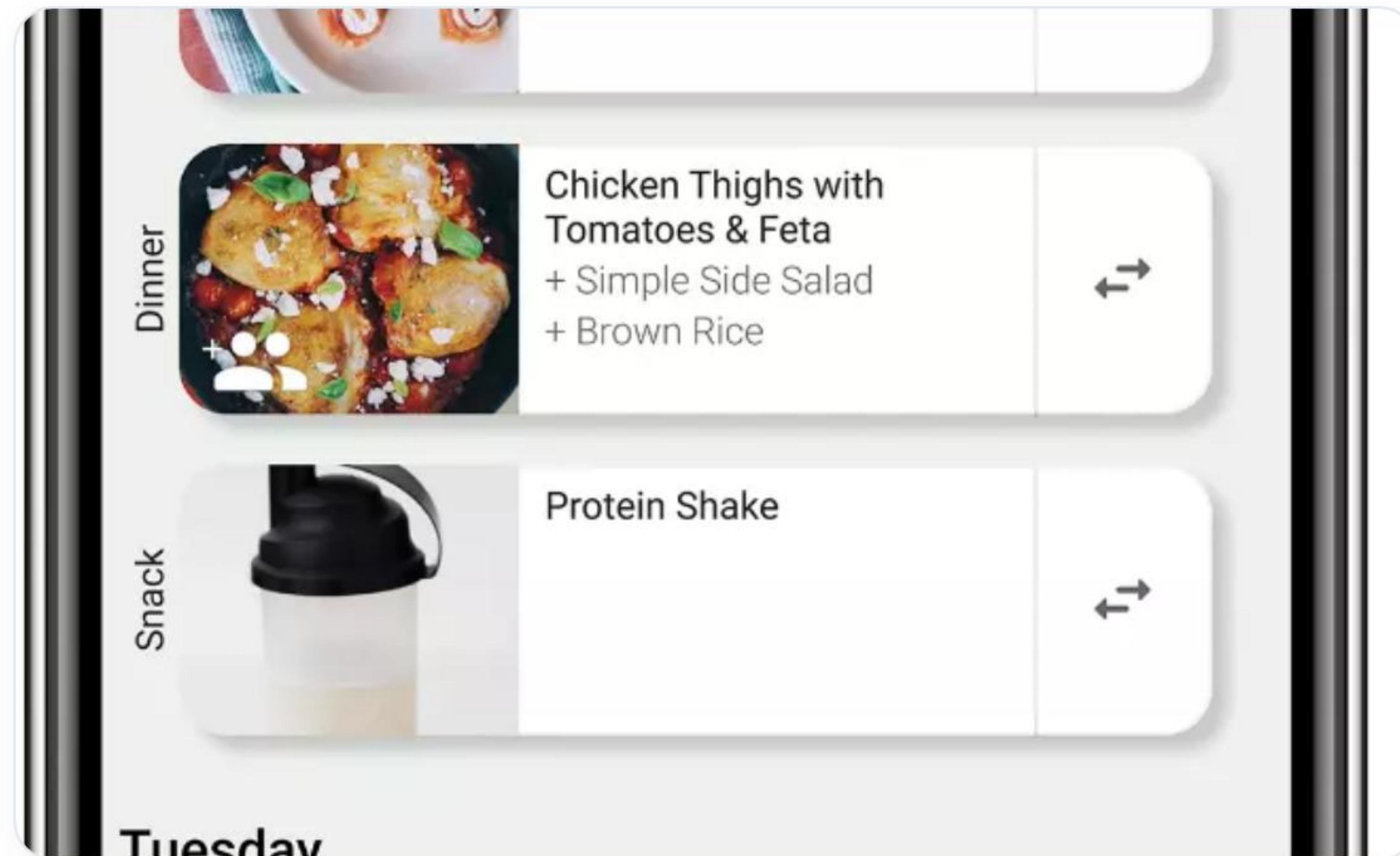
Spine alignment & hold duration timer.



Jumping Jacks

Hand height & leg expansion sync.

Integrated AI Dietician & Personalized Nutrition



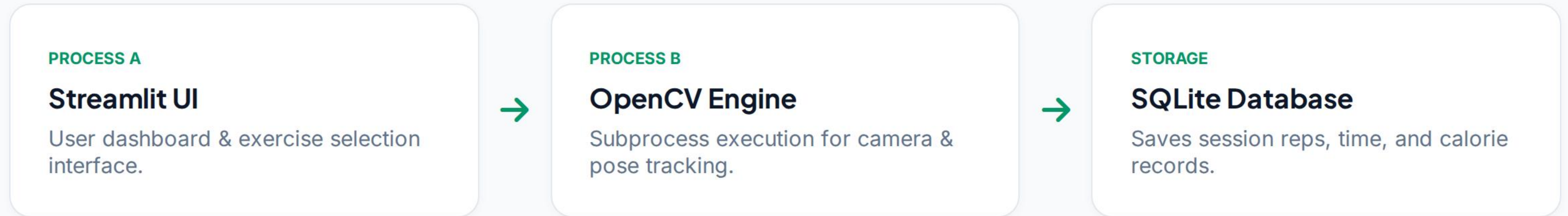
USER QUERY

"What should I eat within 30 minutes post-workout?"

AI DIETICIAN RESPONSE

Consume 25g fast-digesting protein with complex carbs (e.g., whey protein with banana or Greek yogurt with berries).

Multi-Process System Architecture



FEASIBILITY & PRACTICALITY

Zero External Hardware Requirements

100%

Commodity Webcam Compatible

Runs on standard laptop cameras.

30 FPS

Local Edge CPU Inference

No expensive cloud GPU needed.

\$0

Open Source Stack

Python, OpenCV, MediaPipe, SQLite.

Democratizing Personal Health & Fitness Coaching

Injury Reduction

Provides instant feedback on joint angles, stopping dangerous lumbar curve or knee valgus.

Long-term Habit Formation

Gamified milestone balloons and automated session tracking boost user consistency.

FUTURE RELEVANCE

Product Expansion Roadmap

PHASE 1

Voice Coaching

Real-time text-to-speech audio feedback during exercises.

PHASE 2

Mobile & AR App

iOS/Android deployment with AR skeletal overlays.

PHASE 3

Wearable Sync

Heart rate & telemetry integration via Apple Health & Fitbit.



AI Fitness Assistant

Making intelligent, safe, and personalized personal training
accessible to everyone, everywhere.

[Thank You! • Questions & Discussion](#)